COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	September 2, 2026	**DATE SUBMITTED:**	September 3, 2026
NAME:	Group 7	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. _7.1_ Using Sorting Algorithms

1. PROCEDURE OUTPUT

Members:

- Silao, Johrel Louie T.
- Subalisid, Jappeth Royce B.
- Villaruel, Prince Benjamin S.
- Mencidor, Renverlanz D.

Answer the following questions:

1. What is sorting algorithm?

- A process of organizing elements from a collection in an orderly manner. Sorting a large number of items can take a substantial amount of computing resources. In order to sort a collection, it is necessary to compare two values to see which is smaller (or larger) and to have a systematic way to compare values to see if they are out of order. When the values are not in the correct position with respect to one another, it may be necessary to exchange them, with this the total number of comparisons will be the most common way to measure a sort procedure.

2.Where can sorting algorithms be used?

- Sorting algorithms power everyday technology across industries. Practically, they enable databases and search engines to process queries efficiently, allow e-commerce sites to filter products by price and ratings, and help financial systems manage transactions. Behind the scenes, they optimize GPS navigation for nearby locations, organizer file systems and emails, and schedule computer processes by priority. In competitive programming and software engineering, sorting skills are fundamental for solving algorithmic problems. The best algorithm depends on your need, quick sort for speed, merge sort for reliability, or heap sort when memory is limited. Making sorting one of the most important algorithmic concepts developers encounter.

3. Explain the different types of sorting algorithms.

Algorithm	Time complexity Best/Worst	Definition	Difference
Bubble Sort	O(n) / $O(n^2)$	Repeatedly compares adjacent elements and swaps them if they're in the wrong order. Largest	Simplest to understand but the slowest. Ideal for educational purposes and nearly sorted data only.

		elements “bubble” to the end after each pass.	
Selection Sort	$O(n^2)$ / $O(n^2)$	Divides array into sorted and unsorted regions, repeatedly finds minimum element from unsorted region and moves it to sorted region.	Minimizes swaps compared to bubble sort. Better for small datasets where memory writes are expensive.
Insertion Sort	O(n) / $O(n^2)$	Builds sorted array one element at a time by inserting each new element into its correct position within the already sorted portion.	Most efficient for small and nearly sorted datasets, stable algorithm, used in hybrid sorting methods like Timsort.
Merge Sort	O(n log n) / O(n log n)	Divides array into halves recursively, sorts each half, then merges them back together in sorted order.	Guarantees consistent O(n log n) performance. Requires O(n) extra space, ideal for large datasets and external sorting.
Quick Sort	O(n log n) / $O(n^2)$	Selects a pivot element and partitions array into smaller and larger elements, then recursively sorts each partition.	Excellent average-case performance. Generally faster than merge sort in practice, space efficient but unstable.
Heap Sort	O(n log n) / O(n log n)	Builds a max heap from the array and repeatedly extracts the maximum element, placing it at the end of the sorted region.	Guaranteed O(n log n) with O(1) space, slower than quick sort in practice, useful when space is limited.

4. Give sample programs in C++ that use sorting algorithms, specifically selection sort, insertion sort, and bubble sort. Explain how the programs work.

- A book inventory program could use all three sorting algorithms based on what the user desires. If they want to find the top ten shortest books in the library, they could use a selection sort. It would find the shortest book and put them in the first index of the array, then move on to the next shortest, until it finds ten to give to the user. If they wanted to insert new books into an already sorted array, they would use the insertion sort to move the new books one by one to the correct position. Lastly, they could use the bubble sort if they wanted to fix a few unsorted parts of the library, since it looks at adjacent parts, it would fix small mistakes quicker than the other sorting algorithms.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.